

JOBNEX

Find Work Nearby

PROJECT DOCUMENTATION

A digital platform designed to connect workers with nearby shops and businesses that need reliable workforce.

Technology-focused academic project

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Document purpose: This document presents the concept, architecture, modules, technology stack, database design, authentication approach, and future scope of the JobNex platform.

1. Introduction

JobNex — Find Work Nearby is a digital employment platform intended to connect workers with nearby shops, businesses, and local employers. The project focuses on making local job discovery simpler, faster, and more accessible, especially for people looking for part-time, full-time, temporary, or skill-based work.

Traditional local hiring often depends on word of mouth, printed notices, personal contacts, or physically visiting multiple shops. These methods can make it difficult for workers to discover opportunities and for shop owners to find suitable candidates quickly.

JobNex provides a centralized platform where workers can create profiles, search available jobs, view job information, apply for suitable positions, save opportunities, receive notifications, and communicate with shop owners. Shop owners can create shop profiles, publish vacancies, review applications, and communicate with potential workers.

Project Vision

The vision of JobNex is to build a trusted local employment ecosystem where a worker can discover relevant work opportunities nearby and a shop owner can reach suitable workers without relying entirely on informal hiring methods.

Project Motto: **Find the right work. Right near you.**

2. Problem Statement

Local workers and small businesses frequently face difficulties in finding each other efficiently. Workers may not know which nearby shops are hiring, while shop owners may struggle to reach suitable candidates within a short time.

- Limited visibility of local job opportunities.
- Dependence on personal references and offline communication.
- Difficulty comparing job roles, salaries, timings, and locations.
- Time-consuming hiring for small shops and local businesses.
- Lack of a dedicated platform focused on nearby and practical work opportunities.
- Fragmented communication between applicants and employers.

3. Objectives

- Create a user-friendly platform for discovering nearby work opportunities.
- Allow workers to maintain professional profiles containing skills and experience.
- Allow shop owners to create business profiles and publish job vacancies.
- Provide clear job information such as salary, timing, category, description, and location.
- Provide a structured application process for workers.
- Enable communication between workers and shop owners through an in-app chat module.

- Use a reliable cloud database and authentication service through Supabase.
- Build the system with a scalable architecture that can support future mobile applications and advanced services.

4. Proposed Solution

JobNex proposes a centralized web-based employment platform with two primary user groups: **Workers** and **Shop Owners**. The platform combines authentication, profile management, job posting, job search, applications, notifications, saved jobs, and communication.

Basic Workflow

Step	Worker Flow	Shop Owner Flow
1	Register / Login	Register / Login
2	Create profile	Create shop profile
3	Search jobs	Post a job
4	View job details	Receive applications
5	Apply for job	Review applicants
6	Chat with owner	Chat with worker
7	Track applications	Manage job status

5. Features

Feature	Description
User Registration	Account creation for workers and shop owners.
Login & Authentication	Secure sign-in using Supabase Authentication.
Worker Profile	Skills, experience, profession, resume/profile information, and verification details.
Shop Profile	Shop name, category, location, profile information, and active jobs.
Job Search	Search and discover jobs by category and other useful criteria.
Job Details	Salary, timing, location, description, shop information, and application action.
Job Applications	Workers can submit applications and track application status.
Saved Jobs	Workers can save interesting opportunities for later.
Notifications	Updates related to applications, jobs, and platform activity.
Chat	Worker-to-shop-owner communication.
Ratings & Reviews	Future-ready trust and reputation functionality.
Responsive UI	Interface designed for desktop and mobile screen sizes.

6. System Architecture

JobNex follows a modular client–backend architecture. The frontend provides the user interface and communicates with Supabase services through the application service layer. Supabase provides authentication, PostgreSQL database services, storage, and real-time capabilities where required.

Layer	Main Components
Presentation Layer	React components, pages, responsive CSS, navigation, forms, cards, dashboards.
Application Layer	Routing, state management, validation, API/service calls, business rules.
Backend Services	Supabase Authentication, PostgreSQL database, Storage, Realtime services.
External Services	Future integrations such as maps, notifications, AI recommendations, and other APIs.
Users	Workers and Shop Owners interacting through web/mobile interfaces.

Architecture flow: User → JobNex Frontend → Application/Service Layer → Supabase → Database / Authentication / Storage / Realtime

7. Technology Stack

Technology	Purpose
React	Component-based frontend development.
JavaScript	Application logic and interaction.
HTML	Application structure and semantic content.
CSS	Responsive and modern visual design.
Vite	Fast frontend development and build tooling.
React Router	Client-side navigation between application pages.
Supabase	Backend platform for authentication, database, storage, and realtime features.
PostgreSQL	Relational database used by Supabase.
Git / GitHub	Version control and project collaboration.

8. Database Design

The database is designed around relational entities representing users, workers, shop owners, shops, jobs, applications, messages, notifications, saved jobs, and reviews. A relational structure helps maintain data consistency and supports queries across users, jobs, and applications.

Table / Entity	Important Fields	Purpose
profiles	id, role, name, email, phone	Common user information.
workers	id, profile_id, skills, experience, profession	Worker-specific profile data.
shop_owners	id, profile_id	Shop-owner information.
shops	id, owner_id, name, category, address, location	Business/shop information.

Table / Entity	Important Fields	Purpose
jobs	id, shop_id, title, description, salary, timing, status	Published job vacancies.
applications	id, job_id, worker_id, status, applied_at	Worker applications.
messages	id, sender_id, receiver_id, message, created_at	Chat communication.
notifications	id, user_id, title, message, read	User notifications.
saved_jobs	id, worker_id, job_id	Saved job relationships.
reviews	id, reviewer_id, target_id, rating, comment	Ratings and feedback.

Relationship summary: One shop owner can manage one or more shops; a shop can publish multiple jobs; a job can receive multiple applications; a worker can submit multiple applications and save multiple jobs.

9. User Roles

Role	Responsibilities / Access
Worker	Create profile, search jobs, view details, apply, save jobs, receive notifications, chat, and manage profile.
Shop Owner	Create shop profile, post jobs, manage vacancies, view applications, communicate with workers, and manage shop.
Administrator (Future)	Manage users, moderate jobs, handle reports, monitor platform activity, and maintain system policies.

10. Worker Module

The Worker Module is designed for people searching for local employment opportunities. The module focuses on profile quality, job discovery, application management, and communication.

- Worker registration and account verification.
- Profile creation with profession, skills, experience, and other relevant information.
- Search and browse jobs by category.
- View salary, timing, location, description, and shop information.
- Apply to suitable jobs.
- Save jobs for later.
- View application history and status.
- Receive relevant notifications.
- Communicate with shop owners through chat.

11. Shop Owner Module

The Shop Owner Module helps local businesses advertise vacancies and manage applicants from a centralized interface.

- Shop owner registration and authentication.
- Shop profile creation and management.

- Post new jobs with title, category, description, salary, timing, and location.
- View active and previously posted jobs.
- Receive and review worker applications.
- Communicate with applicants through chat.
- Update job status when a position is filled or closed.
- Maintain shop information and reputation features.

12. Job Module

The Job Module is the core discovery component of JobNex. Each job should contain enough information for a worker to understand the opportunity before applying.

Field	Example / Description
Job Title	Delivery Worker, Mechanic, Helper, Electrician, etc.
Category	Delivery, Helper, Cleaning, Hotel, Mechanic, Electrician, etc.
Salary	Expected salary or pay range.
Timing	Working hours or shift details.
Location	Shop or workplace location.
Description	Responsibilities and requirements.
Status	Active, paused, filled, or closed.
Shop	Associated business profile.

13. Application Module

The Application Module manages the relationship between a worker and a job vacancy. A worker can apply to an active job, while the shop owner can review and update the application status.

- Application creation when a worker applies.
- Prevention of inappropriate duplicate submissions through database rules and application logic.
- Application status such as Pending, Shortlisted, Accepted, or Rejected.
- Worker-side application history.
- Shop-owner-side applicant list.
- Notification support for important status changes.

14. Chat Module

The Chat Module provides a direct communication channel between workers and shop owners. It is intended to reduce dependence on external messaging platforms during the hiring process.

- One-to-one conversations between authorized users.
- Message text and timestamps.

- Conversation history stored securely in the database.
- Unread-message indicators.
- Realtime message updates using Supabase Realtime where enabled.
- Future support for attachments and richer communication features subject to security and storage policies.

15. Authentication

Authentication verifies user identity before access to protected application functionality. JobNex can use Supabase Authentication to manage account creation, login sessions, email verification, password-based authentication, and future OAuth providers.

Authentication Flow

Stage	Process
Registration	User enters required account information and creates an account.
Verification	Email verification can be enabled according to project requirements.
Login	User signs in using registered credentials.
Session	Supabase manages the authenticated session.
Authorization	Application checks user role and protected routes before allowing actions.
Logout	Session is securely terminated on user request.

Security note: Passwords should never be stored manually in application tables. Sensitive credentials and service-role keys must not be exposed in frontend source code or committed to GitHub.

16. Supabase Integration

Supabase acts as the backend platform for JobNex. It provides PostgreSQL database functionality together with authentication, storage, and realtime capabilities.

Service	JobNex Usage
Authentication	Registration, login, sessions, email verification, and identity management.
PostgreSQL Database	Users, profiles, shops, jobs, applications, messages, notifications, saved jobs, and reviews.
Storage	Profile images, shop images, resumes, and other permitted files.
Realtime	Chat updates and selected live notifications.
Row Level Security	Restrict access so users can only perform permitted database operations.

Recommended environment variables: **VITE_SUPABASE_URL** and **VITE_SUPABASE_ANON_KEY**. Keep local environment files such as `.env` outside version control.

17. Screenshots

The final project repository should contain screenshots demonstrating the implemented interface. Recommended screenshots include the home page, job listing, job details, dashboard, profile, registration/login, and other completed modules.

Screenshot	Recommended Content
Screenshot 1	JobNex home / landing page.
Screenshot 2	Job listing or job details page.
Additional	Dashboard, login/register, profile, applications, chat, and settings as available.

SCREENSHOT PLACEHOLDER

Insert the final JobNex UI screenshot here when preparing the printed/submitted version.

18. Future Scope

JobNex can be expanded from a web project into a larger local employment ecosystem. The following enhancements can improve usability, trust, discovery, and scalability.

Enhancement	Potential Benefit
Google Maps / Location Services	Better nearby-job discovery and navigation.
Mobile Application	Dedicated Android/iOS experience.
AI Recommendations	Suggest jobs based on skills, experience, category, and preferences.
Voice Search	Simpler job discovery for users who prefer voice interaction.
Multi-language Support	Improved accessibility for regional users.
QR Attendance	Digital attendance for participating workplaces.
Online Payments	Future support for secure digital payment workflows.
Video Interview	Remote screening for selected jobs.
Advanced Admin Panel	Moderation, reports, analytics, and platform management.
Ratings & Verification	Improve trust between workers and employers.
Analytics	Insights into job demand, applications, and platform activity.

19. Conclusion

JobNex — Find Work Nearby is designed to address a practical local employment problem by bringing workers and small businesses together through a single digital platform. The project provides the foundation for job discovery, worker and shop-owner profiles, vacancy management, applications, notifications, and communication.

Using React for the frontend and Supabase for backend services and PostgreSQL data management provides a modern and scalable starting point. The modular architecture also allows future features such as maps, mobile applications, AI recommendations, multilingual support, verification, and advanced administration to be added without redesigning the complete system.

Final Outcome: JobNex aims to make local hiring more discoverable, organized, and convenient for both workers and shop owners.

— End of Project Documentation —